CHAPTER

5 Chemical Bonding

For NSEJS 2025

INTRODUCTION

It is a well known fact that none of the elements exist as independent atoms, except for noble gases, Most of the elements exist as molecules which are in a cluster. The questions that arise are.

- Why do atoms combine and form bonds ?
- How do atoms combine to form molecules ?
- Why is a molecule more stable than the individual atom ?
- Such doubts will be discussed in this chapter.

DEFINATION:

(i) An attractive force that acts between two or more particles (atoms, ions or molecules) to hold them together, is known as a chemical bond.

(ii) It is a union of two or more atoms to acquire stable inert gas configuration s^2p^6.

To explain nature of chemical bond Different theories are given at first KOSSEL & LEWIS proposed the theory.

Kossel & Lewis approach chemical bonding: In early days, the ability if various elements to combine with one another was expressed in terms of their valency. The concept of valency was not based on any logical understanding. The development in the field of atomic structure provide foundations for various theories of valency. The initial contributions in this field came from. W. Kossel and G. N. Lewis in 1916. They were the first to provide some logical explanation of valency which based on the inertness of noble gases. This view, later on, came to be known as octet rule.

OCTET RULE

(Tendency to acquire noble gas configuration)

(i) It has been observed that atoms of noble gases have little or no tendency to combine with each other or with atoms of other elements.

(ii) It means that these atoms must have a stable electronic configuration.

(iii) These elements (noble gases) have 8 electrons ($ns^2\ np^6$) except helium which has 2 electrons ($1s^2$) in their outer most shell.

Element	**Outer most shell configuration**
Ne	$2s^22p^6$
Ar	$3s^23p^6$
Kr	$4s^24p^6$
Xe	$5s^25p^6$
Rn	$6s^26p^6$

(iv) It is therefore concluded that s^2p^6 configuration in the outer energy level constitutes a structure of maximum stability or minimum energy

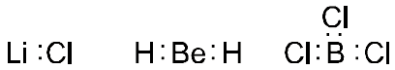
(v) “Tendency of atoms to have eight electrons in their outermost shell is known as Lewis octet rule". To achieve inert gas configuration atoms lose, gain or share electrons.

Limitations of the Octet Rule :

The octet rule, though useful, is not universal. It is quite useful for understanding. The structures of most of the organic compounds and it applies mainly to the second period elements of the periodic table. There are three types of exceptions to the octet rule.

1. The incomplete octet of the central atom

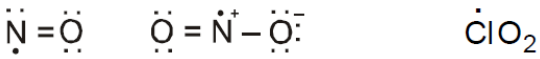
In some compounds, the number of electrons surrounding the central atom Is less than eight. This is especially the case with elements having less than four valence electrons. Examples are LiCl. BeH_2 and BCl_3.



Li. Be and B have 1, 2 and 3 valence electrons only. Some other such compounds are $AlCl_3$ and BF_3.

2. Odd-electron molecules

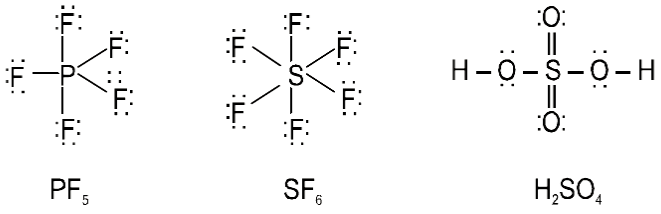
In molecules with an odd number of electrons like nitric oxide. NO and nitrogen dioxide. NO_2, the octet rule is not satisfied for all the atoms



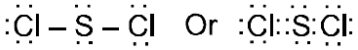
3. The expanded octet/ super octet / hypervalent compound

Elements in and beyond the third period of the periodic table have, apart from 3s and 3p orbitals, 3d orbitals also available for bonding. In a number of compounds of these elements there are more than eight valence electrons around the central atom. This is termed as the expanded octet. Obviously the octet rule does not apply in such cases.

Some of the examples of such compounds are: PF_5 SF_6, H_2SO_4 and a number of coordination compounds.



Interestingly, sulphur also forms many compounds in which the octet rule is obeyed. In sulphur dichloride, the S atom has an octet of electrons around it.

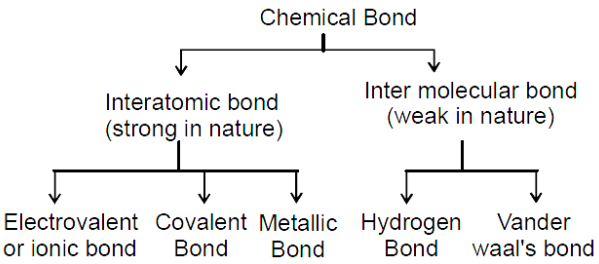


4. Other drawbacks of the octet theory

- It is clear that octet rule is based upon the chemical Inertness of noble gases. However, some noble gases (for example xenon and krypton) also combine with oxygen and fluorine to form a number of compounds like XeF_2, KrF_2, $XeOF_2$ etc.,
- This theory does not account for the shape of molecules.
- It does not explain the relative stability of the molecules being totally silent about the energy of a molecule.

CLASSIFICATION OF CHEMICAL BONDS:

On the basis of electronic valency theory and structure, chemical bonds can be classified as follows.



Bond	Bond Energy per mole
Ionic, Covalent & Co–ordinate	200 kJ
Hydrogen bonding	10 – 40 kJ
Vander–waal's	2 – 10 kJ

ELECTROVALENT OR IONIC BOND

The chemical bond formed between two or more atoms as a result of complete transfer of one or more electrons from one atom to another is called ionic or electrovalent bond.

Electrovalent bond is not possible between similar atoms. This type of bonding requires two atoms of different nature. One atom should have the tendency to loose electrons i.e. electropositive in nature and the other atom should have the tendency to accept electrons i.e. electronegative in nature.

Electropositive atom looses electrons (group IA to IIIA)

Electronegative atom gains electron (group IVA to VII A)

Example : IA and VII A group elements form strong ionic compound.

$Na + Cl \longrightarrow Na^{+} + Cl^{-}$

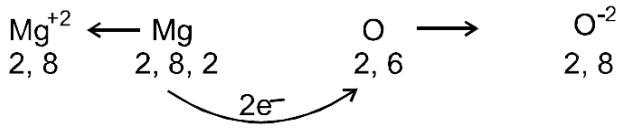
2, 8, 1 2, 8, 7 2, 8 2, 8, 8

More the distance between two elements in the periodic table more will be the ionic character of the bond.

Total number of electrons lost or gained is called electrovalency.

Example :

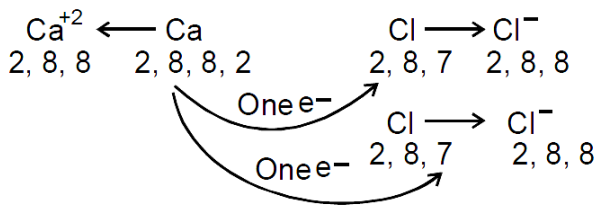
(A) $Mg + O \longrightarrow MgO$



electrovalency of Mg = 2

electrovalency of O = 2

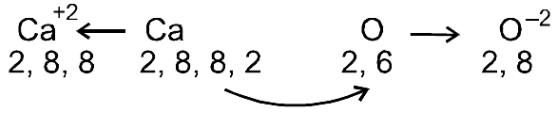
(B) $Ca + 2Cl \longrightarrow CaCl_2$



electrovalency of Ca = 2

electrovalency of Cl = 1

(C) $Ca + O \longrightarrow CaO$



electrovalency of Ca = 2

electrovalency of O = 2

Electrostatic force of attraction between cation and anion is called ionic bond or electrovalent bond.

Force of attraction is equal in all direction so ionic bond is non–directional.

A definite three dimensional structure called crystal lattice is formed.

Energy released during the formation of one mole crystal lattice is called lattice energy.

e.g. $(6.023 \times 10^{23})Na^+ + (6.023 \times 10^{23})\ Cl^- \longrightarrow$

(6.023×10^{23}) NaCl + 94.5 K.cal (Lattice energy).

Ionic compounds do not have a molecular formula, they only have empirical formula.

e.g. NaCl is the empirical formula of sodium chloride.

Factors favouring formation of Ionic bonds :

Formation of ionic bond depends upon three factors :

(A) Ionisation energy (IE) :

Amount of energy required to remove an electron from the outermost orbit of an isolated gaseous atom to form positive ion or cation is called ionization energy [energy is absorbed so it is an endothermic process]

$M + I.E. \longrightarrow M^+ + e^-$

Less Ionisation energy ⇒ Greater tendency to form cation.

Example :

$Na^+ > Mg^{+2} > Al^{+3}$

$Cs^+ > Rb^+ > K^+ > Na^+ > Li^+$

} Cation formation tendency

(B) Electron affinity :

Amount of energy released when an electron is added to an isolated gaseous atom to form negative ion or anion is called electron affinity [energy is released so it is an exothermic process]

$X + e^- \longrightarrow X^- + EA$

High electron affinity ⇒ Greater tendency to form anions

$Cl^- > F^- > Br^- > I^-$

$F^- > O^{-2} > N^{-3}$

} Anion formation tendency

Characteristics of ionic compounds :

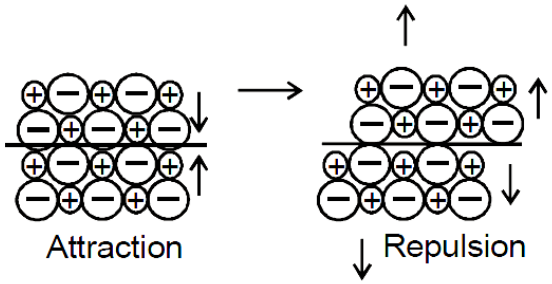
(i) Physical state

(a) Electrovalent compounds are generally crystalline, hard & brittle in nature.

(b) These compounds are generally made from ions which are arranged in a regular way as a lattice structure.

(c) Thus electrovalent compounds exist as three dimensional solid aggregates.

(d) Normally each ion is surrounded by a number of oppositively charged ions and this number is called coordination number



{Same charged ions comes near.
So they repell each other}

(ii) Boiling point and melting point :

High boiling point and melting points are due to strong electrostatic force of attraction.

(iii) Electrical conductivity :

It depends on ionic mobility. In solid state there are no free ions so they are bad conductors of electricity

In fused state or aqueous solution free ions are present so they are good conductors of electricity (Conductivity order) Solid state < Fused state < Aqueous solution

(iv) Ionic reactions :

☞ Ionic compounds show ionic reactions & covalent compounds show molecular reactions.

Ionic reactions are fast reactions.

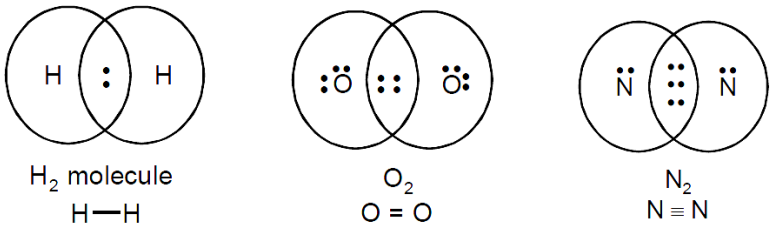
Example : When NaCl is added to $AgNO_3$ solution, white ppt of AgCl is formed at once.

$$Ag^+NO_3^- + Na^+Cl \rightleftharpoons Na^+NO_3^- + AgCl\downarrow$$

white ppt.

COVALENT BOND

A covalent bond may be defined as the bond formed by mutual sharing of electrons between the participating atoms (which are short of electrons) of comparable electronegativity.



H_2 molecule
H—H

O_2
O = O

N_2
N ≡ N

Covalency :

(i) It is defined as the number of electrons contributed by an atom of the element for sharing with other atom so as to achieve noble gas configuration.

(ii) It can also be defined as the number of covalent bonds formed by the atom of the element with other atoms.

(iii) The usual covalency of an element except hydrogen is equal to 8 minus the number of group to which the element belongs.

Lewis structure and covalent bond :

(i) Structures in which valency electrons are represented by dots are called Lewis structures.

(ii) All atoms in the formulae will have eight electrons in it's valence shell. H atom is an exception for it can form large number of bonds with the atom present at the center of any crystal structure. Other atoms surround it to complete their octet.

(iii) Lewis dot formulae shows only the number of valency electrons, the number and kinds of bonds, but does not depict the three dimensional shapes of molecules and polyatomic ions.

(iv) Lewis formulae are based on the fact that the representative elements achieve a noble gas configuration in most of their compounds i.e. 8 electrons in their outermost shell (except for H_2, Li^+ and Be^{2+} ions which have only 2 electrons)

Characteristics of covalent compounds :

(i) Physical state :

(a) Under normal temperature and pressure, they exist as gases or liquids of low boiling points.

(b) This is due to the fact that very weak forces of attraction (Vander waal's forces) exist between the molecules due to which molecules are far from each other.

(c) If their molecular masses are high they exist as soft solids ex. Sulphur, phosphorus and iodine

(ii) Crystal structures :

Various types of structures that are present in a covalent compound are as follows.

(a) Covalent solid : In this type of structure every atom is bonded to four other atoms by single covalent bonds resulting in the formation of a giant structure e.g. SiC, AIN and diamond these crystals are very hard and possess high melting point.

Diamond is sp^3 hybridized carbon atom and it forms a tetrahedral structure.

Graphite is sp^2 hybridized carbon atoms and it forms hexagonal layers which can slide over each other due to weak Vander waal's forces of attraction. Distance between C–C atom is 1.42 Å distance between layers is 3.6 Å. **Graphite is more stable than diamond due to high value of change in enthalpy.**

(b) Molecular solids : They are formed when one atom combines with another by a covalent bond and then the molecule combines with another similar molecule with the help of Vander waal's force of attraction or hydrogen bond

Example : CH_4(Solid), dry ice, ice

(iii) Melting and boiling points :

(a) With the exception of a few which have giant three dimensional structures such as diamond, carborundum (SiC), Silica (SiO_2), others have relatively low melting and boiling points.

(b) This is due to the presence of weak attractive forces between the molecules.

(iv) Electrical conductivity :

(a) In general covalent substances are bad conductors of electricity. Since they do not contain charged particles or free electrons.

(b) Substances which have polar character like HCl in a solution, can conduct electricity.

(c) Graphite can conduct electricity since electrons can pass from one layer to other.

(d) Some show conductivity due to self ionisation. example Liq NH_3

$$NH_3 + NH_3 \longrightarrow NH_4^+ + NH_2^-$$

$$H_2O + H_2O \longrightarrow H_3O^+ + OH^-$$

(v) Chemical reactions :

Covalent substances give molecular reactions. Reaction rate is usually low because it involves two steps

(i) breaking of covalent bonds of the reactants (ii) establishing new bonds. While in ionic reactions there is only regrouping of ions.

(vi) Solubility :

(a) Non polar compounds are soluble in non polar solvents. Non polar solvents are CCl_4, Benzene, CS_2 etc.

(b) Polar compounds are soluble in polar solvents (i.e. ΔEN 0.9 to 1.8). Polar solvents are H_2O, $CHCl_3$, Alcohol etc.

(vii) Isomerism :

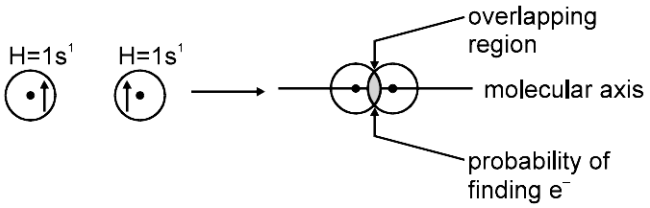
Covalent bonds are rigid and directional. On account of this there is a possibility of different arrangement of atoms in space. Covalent compounds can thus shows isomerism (structural and spatial)

ORBITAL OVERLAP CONCEPT

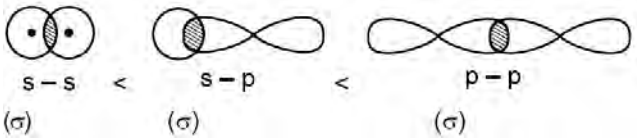
In the formation of hydrogen molecule, there is a minimum energy state when two hydrogen atoms are so near that their atomic orbitals undergo partial interpenetration. This partial merging of atomic orbitals is called overlapping of atomic orbitals which results in the pairing of electrons. The extent of overlap decides the strength of a covalent bond. In general, greater the overlap the stronger is the bond formed between two atoms. Therefore, according to orbital overlap concept, the formation of a covalent bond between two atoms results by pairing of electrons present in the valence shell having opposite spins.

Main points of valency bond theory:

(i) A covalent bond is formed by partial overlapping of two atomic orbitals



(ii) More is the extent of overlapping between the two atomic orbital, stronger will be bond.



[Principal Quantum no. same, n = 2]

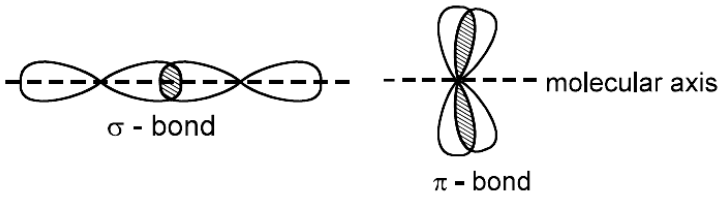
∵ s orbital are spherical in nature so they are least diffused hence it will provide less area for overlapping.

(iii) Orbitals which are undergoing overlapping must be such that

(a) Each orbital should have one electron with opposite spin (for formation of covalent bond)

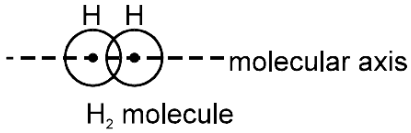
(b) One orbital have pair of electron and the other orbital have no electron (for formation of co-ordinate bond)

(iv) If the overlapping is along the molecular axis then bond will be sigma (σ) & in the perpendicular direction, it will be pi(π) bond.

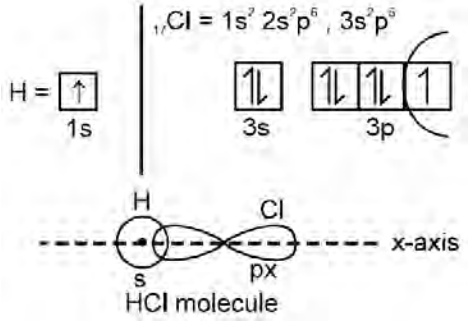


☞ **Examples of overlapping of pure atomic orbitals.**

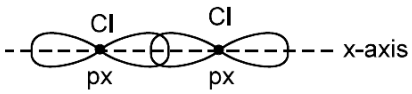
(i) H_2 (s–s) H = ↑ (1s) H = ↑ (1s)



(ii) HCl gas molecule (s-p)



(iii) F_2, Cl_2, Br_2, I_2 (p-p)



F_2 Cl_2 Br_2 I_2

2p-2p 3p-3p 4p-4p 5p-5p (Type of overlapped orbitals)

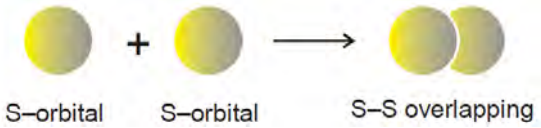
Types of Overlapping and Nature of Covalent Bonds :

The covalent bond may be classified into two types depending upon the types of overlapping :

(i) Sigma (σ) bond, and (ii) pi (π) bond

(i) **Sigma (σ) bond :** This type of covalent bond is formed by the end to end (hand-on) overlap of bonding orbitals along the intemuclear axis. This is called as head on overlap or axial overlap. This can be formed by any one of the following types of combinations of atomic orbitals.

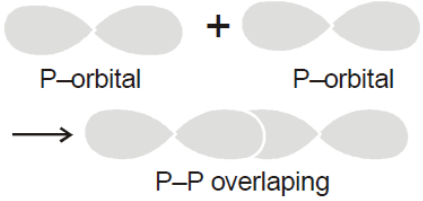
- **s-s overlapping :** In this case, there is overlap of two half filled s-orbitals along the internuclear axis as shown below :



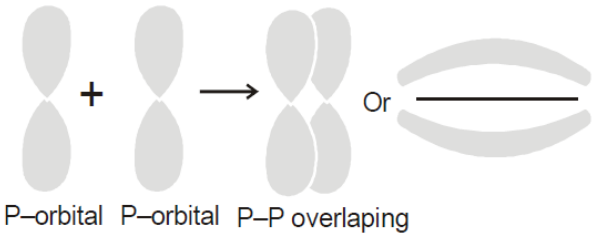
- **s-p overlapping :** This type of overlap occurs between half filled s-orbitals of one atom and half filled p-orbitals of another atom.



- **p-p overlapping :** This type of overlap takes place between half filled p-orbitals of the two approaching atoms.



(iii) **pi(π) bond :** In the formation of π bond the atomic orbitals overlap in such a way that their axes remain parallel to each other and perpendicular to the internuclear axis. The orbitals formed due to sidewise overlapping consists of two saucer type charged clouds above and below the plane of the participating atoms.



Strength of Sigma and pi Bonds :

Basically the strength of a bond depends upon the extent of overlapping- In case of sigma bond, the overlapping of orbitals takes place to a larger extent. Hence, it is stronger as compared to the pi bond where the extent of overlapping occurs to a smaller extent. Further, it is important to Mind it that pi bond. between two atoms is formed in addition to a sigma bond. It is always present in the molecules containing multiple bond (double or triple bonds)

Difference in σ and π bonds :

S.No	Sigma (σ) bond	Pi (π) bond
1.	It results from the end to end overlapping of two σ orbitals or two p–orbitals or one s and one p–orbital.	It result from the sidewise (lateral) overlapping of two p orbitals.
2.	Its bonded orbital consists of a single electron cloud symmetrical about internuclear Axis	Its bonded orbital consists of two electron clouds one above and the other below the plane of symmetry
3.	Strong	Weak
4.	Bond energy 80 Kcals	65 Kcals
5.	More stable	Less Stable
6.	Less reactive	More reactive
7.	Can independently exist	Always exists along with a σ bond hybridsation
8.	Hybridization depends upon s bond	doesn't depend on s bond
9.	The groups or atoms can undergo bond rotation about single sigma (σ) bonds	Due to resistance to rotation around the π bond the groups attached to it are not free to rotate.
10.	The s electrons are referred as localised	in π bond the electrons are held less firmly bond thus can be easily dissociated or polarised by an external charge and hence the πe^- are referred as mobile electrons.
11.	Shape of the molecule is determined by the σ bonds present in the molecule	π bonding does not affect the shape of the molecule

HYBRIDISATION

The valence bond theory (overlapping concept) explains satisfactorily the formation of various molecules but it fails to account the geometry and shapes of various molecules. It does not give the explanation why $BeCl_2$ is linear, BF_3 is planar, CH_4 is tetrahedral, NH_3 is pyramidal and water is V– shaped molecule.

In order to explain these cases, the valence bond theory has been supplemented by the concept of hybridization. This is a hypothetical concept and was introduced by Pauling & Slater.

According to them the atomic orbitals combine to form new set of equivalent orbitals known as **hybrid orbitals.**

Unlike pure orbitals, the hybrid orbitals are used in bond formation. The phenomenon is known as **hybridisation** which can be defined as the process of intermixing of the orbitals of slightly different energies in the formation of new set of orbitals of equivalent energies and shape. For example when one 2s and three 2p-orbitals of carbon hybridise, there is the formation of four new sp^3 hybrid orbitals.

Salient features of hybridisation : The main features of hybridisation are as under :

1. The number of hybrid orbitals is equal to the number of the atomic orbitals that get hybridised.

2. The hybridised orbitals are always equivalent in energy and shape.

3. The hybrid orbitals are more effective in forming stable bonds than the pure atomic orbitals.

4. These hybrid orbitals are directed in space in some preferred direction to have minimum repulsion between electron pairs and thus a stable arrangement is obtained. Therefore, the type of hybridisation indicates the geometry of the molecules.

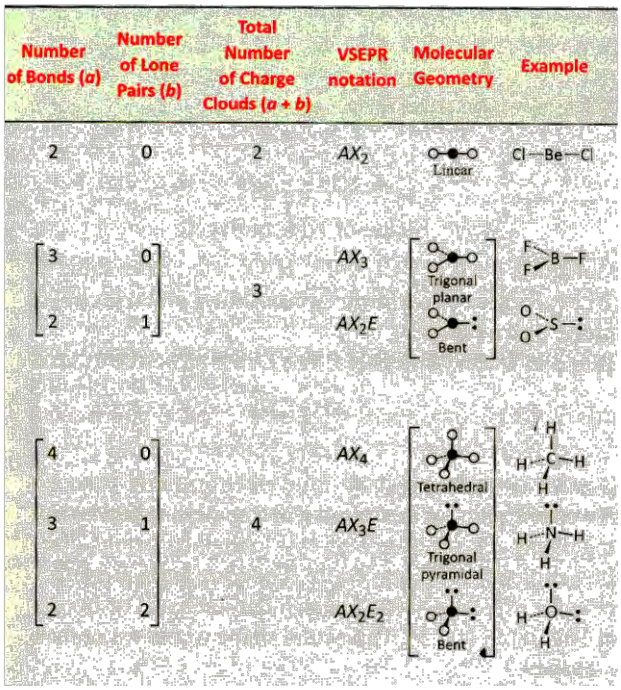
Important conditions for hybridisation :

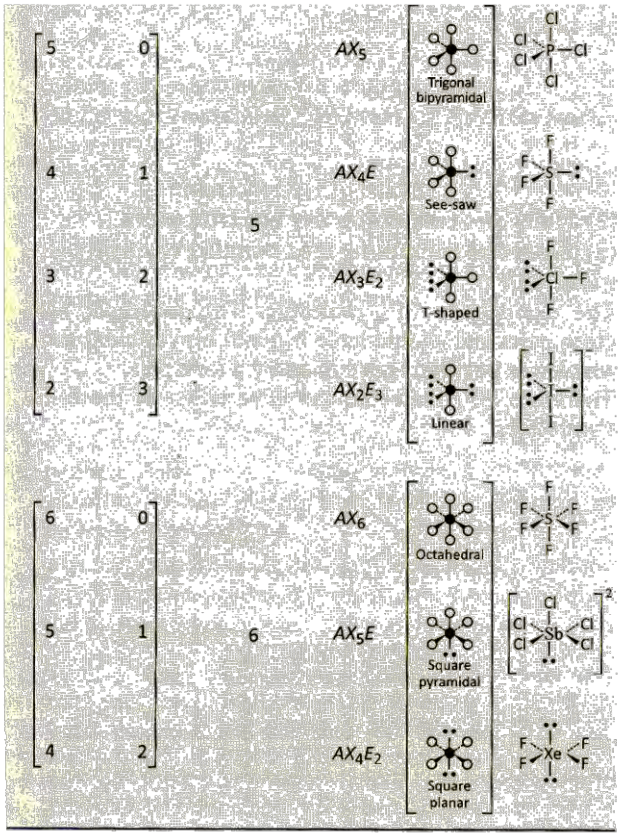
(i) The orbitals present in the valence shell of the atom are hybridised.

(ii) The orbitals undergoing hybridisation should have almost equal energy.

(iii) Promotion of electron is not essential condition prior to hybridisation.

(iv) It is the orbital that undergo hybridization and not the electrons. For example, for orbitals of nitrogen atom ($2s^2\ 2p^1_x\ 2p^1_y\ 2p^1_z$) belonging to valency shell when hybridize to form four hybrid orbitals, one of which has two electrons (as before) and other three have one electron each. It is not necessary that only half filled orbitals participate in hybridisation. In some cases, even filled orbitals of valence shell ake part in hybridisation.





Determination of hybridisation of an atom in a molecule or ion:

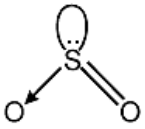
Steric number rule (given by Gillespie) :

Steric No. of an atom = number of atom bonded with that atom + number of lone pair(s) left on that atom.

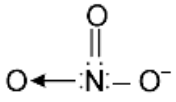
Mind it : This rule is not applicable to molecules/ions which have odd e^- (ClO_2, NO, NO_2), free radicals and compounds like B_2H_6 which involve 3 centre $2e^-$ bond (banana bond).

For example : O = C = O S.No. = 2 + 0 = 2

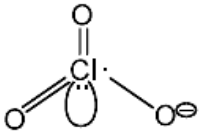
S.No. = 2 + 1 = 3



S.No. = 3 + 0 = 3



S.No. = 3 + 1 = 4



Steric number	Types of Hybridisation	Geometry
2	sp	Linear
3	sp^2	Trigonal planar
4	sp^3	Tetrahedral
5	$sp^3 d$	Trigonal bipyramidal
6	$sp^3 d_2$	Octahedral
7	$sp^3 d_3$	Pentagonal bipyramidal

(i) sp hybridisation :

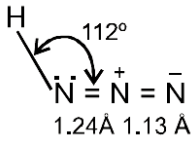
Examples :

(a) —C ≡ H – C ≡ N, H – C ≡ C – H

(b) = C = O = C = O $H_2C = C = CH_2$

(c) N_3^- (azide ion), BeF_2, $HgCl_2$, NO_2^+ (nitronium ion), N_2O

Mind it : In N_3^- there is a total of two σ and two π bonds giving N = N = N and a bond order of two. Both N – N bonds are the same length, 1.16 Å. The hydrogen azide molecules has a bent structure and two N – N bond length are different.



(ii) sp^2 hybridisation :

Examples

Type	AB_3	AB_2L
Shape	Trigonal	V-shape (bent)
Example	$>C=$, CH_3, CH_3 C_6H_6, CO_3^{2-}, HCO_3^- H_2CO_3, graphite, BF_3 $B(OH)_3$, SO_3, NO_3^- C_{60} (Fullerene)	$-\ddot{C}-$, SO_2, $SnCl_2$ NOCl, O_3, NO_2 NO_2^-

Where A = central atom, B = side atom, L = lone pair of e^-

(iii) sp^3 hybridisation :

Type	AB_4	AB_3L	AB_2L_2	ABL_3
Shape	tetrahedral	pyramidal	V-shape or bent	Linear
Example	CH_4	XeO_3	OBr_2	^-OCl

Hybridization involving d-orbital :

Type of 'd' orbital involved

$sp^3 d$ dz^2

sp^3 d^2 $dx^2 - y^2$ & dz^2

sp^3 d^3 dxy, dyx, dzx

dsp^2 $d_{x^2-y^2}$

(iv) sp^3d hybridiation :

Type	AB_5	AB_4L	AB_3L_2	AB_2L_3
Shape	Trigonal bipyramidal	See-saw	T-shape	Linear
Example	PCl_5, PBr_5, PF_5 etc.	SF_4, XeO_2F_2	ClF_3, $[XeF_3]^+$	XeF_2, N_3^- $[Cl_2]$

(v) sp^3d^2 hybridization :

Type	AB_6	AB_5L	AB_4L_2
Shape	Octahedral	Square Pyramidal	Square Planar
Example	SF_6, PF_6^-, $[SiF_6]^{2-}$, $[AlF_6]^{3-}$, $[XeO_6]^{4-}$	BrF_5, F_5XeOF_4	$Cl_4^-XeF_4$

Solved Examples

Example-1 Which is linear PH_4^+, H_3S^+ or NO_2^+ ?

(A) Phosphonium ion (PH_4^+) (B) Sulphonium (H_3S^+)

(C) Nitronium ion (NO_2^+) (D) None of these

Solution : Phosphonium ion (PH_4^+) is tetrahedral, sulphonium (H_3S^+) is pyramidal, but nitronium ion (NO_2^+) is $O = \overset{\oplus}{N} = O$ linear ; (sp hybridisation).

Example-2 The triodide ion (I_3^-) in which the I atoms are arranged in a straight line is stable, but the corresponding F_3^- ion does not exist.

Solution : F_3^- and I_3^- are of same group. Iodine can expand its octet but F cannot and thus, in I_3^- octet rule is not violated, but in F_3^- octet rule is violated.

INTERMOLECULAR FORCES

Intermolecular forces are the forces of attraction and repulsion between interacting particles (atoms and molecules). This term does not include the electrostatic forces that exist between the two oppositely charged ions and the forces that hold atoms of a molecule together i.e., covalent bonds. Attractive intermolecular forces are known as **van der Waals forces**, in honour of Dutch scientist Johannes van der Waals (1837-1923), who explained the deviation of real gases from the ideal behaviour through these forces. It is important to Not that attractive forces between an ion and a dipole are known as ion-dipole forces and these are not van der Waals forces. We will now learn about There are different types of van der Waals forces.

(i) Dispersion Forces or London Forces

Atoms and non-polar molecules are electrically symmetrical and have no dipole moment because their electronic charge cloud is symmetrically distributed. But a dipole may develop momentarily even in such atoms and molecules. This can be understood as follows.

Suppose we have two atoms 'A' and 'B' in the close vicinity of each other (Fig. a). It may so happen that momentarily electronic charge distribution in one of the atoms, say 'A', becomes unsymmetrical *i.e.,* the charge cloud is more on one side than the other (Fig. b and c). This results in the development of instantaneous dipole on the atom 'A' for a very short time. This instantaneous or transient dipole distorts the electron density of the other atom 'B', which is close to it and as a consequence a dipole is

induced in the atom 'B'. The temporary dipoles of atom 'A' and 'B' attract each other. Similarly temporary dipoles are induced in molecules also.

This force of attraction was first proposed by the German physicist Fritz London, and for this reason force of attraction between two temporary dipoles is known as **London force**. Another name for this force is **dispersion force**. These forces are always attractive and interaction energy is inversely proportional to the sixth power of the distance between two interacting particles (i.e., $1/r^6$ where r is the distance between two particles). These forces are important only at short distances (~500 pm) and their magnitude depends on the polarisability of the particle.

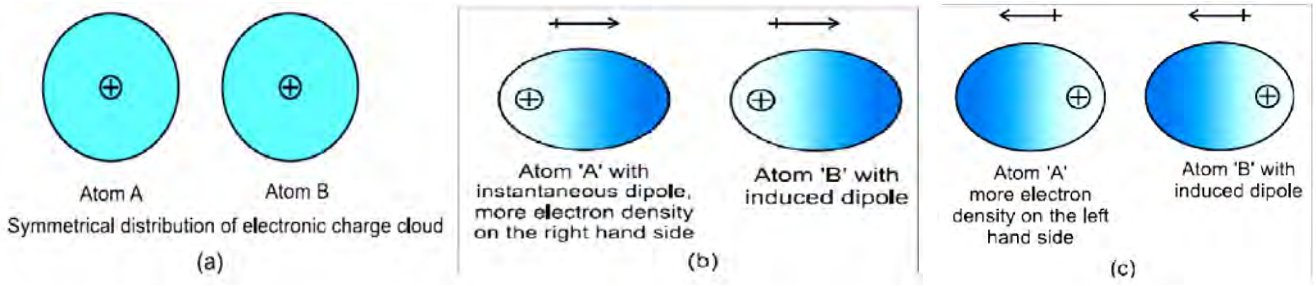


Fig : Dispersion forces or London forces between atoms.

(ii) Dipole-dipole forces

Dipole-dipole forces act between the molecules possessing permanent dipole. Ends of the dipoles possess **partial charges** and these charges are shown by Greek letter *delta* **σ**. Partial charges are always less than the unit electronic charge ($1.6 \sim 10^{-19}$ C). The polar molecules interact with neighbouring molecules. Fig (a) shows electron cloud distribution in the dipole of hydrogen chloride and Fig. (b) shows dipole-dipole interaction between two HCl molecules. This interaction is stronger than the London forces but is weaker than ion-ion interaction because only partial charges are involved. The attractive force decreases with the increase of distance between the dipoles. As in the above case here also, the interaction energy is inversely proportional to distance between polar molecules. Dipole-dipole interaction energy between stationary polar molecules (as in solids) is proportional to $1/r^3$ and that between rotating polar molecules is proportional to $1/r^6$, where r is the distance between polar molecules. Besides dipole-dipole interaction, polar molecules can interact by London forces also. Thus cumulative effect is that the total of intermolecular forces in polar molecules increase.**73**

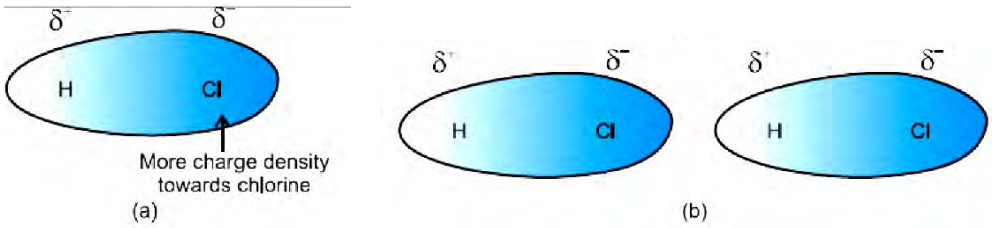


Fig : (a) Distribution of electron cloud in HCl- a polar molecule.
(b) Dipole-dipole interaction between two HCl molecules

(ii) Dipole–Induced Dipole Forces

This type of attractive forces operate between the polar molecules having permanent dipole and the molecules lacking permanent dipole. Permanent dipole of the polar molecule induces dipole on the electrically neutral molecule by deforming its electronic cloud. Thus an induced dipole is developed in the other molecule. In this case also interaction energy is proportional to $1/r^6$ where r is the distance between two molecules. Induced dipole moment depends upon the dipole moment present in the permanent dipole and the polarisability of the electrically neutral molecule. Molecules of larger size can be easily polarized. High polarisability increases the strength of attractive interactions.

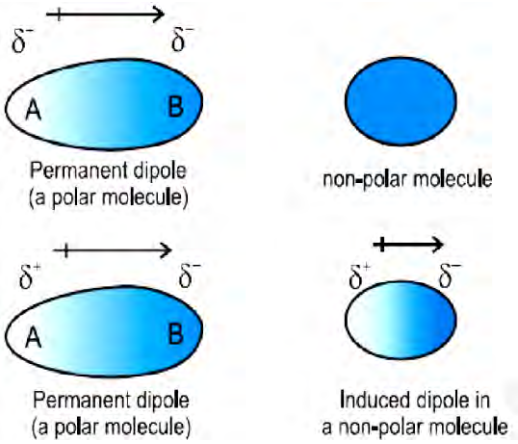


Fig: Dipole - induced dipole interaction between permanent dipole and induced dipole

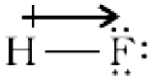
In this case also cumulative effect of dispersion forces and dipole-induced dipole interactions exists.

POLARITY OF BOND

The existence of a hundred percent ionic or covalent bond represents an ideal situation. In reality no bond or a compound is either completely covalent or ionic. Even in case of covalent bond between two hydrogen atoms, there is some ionic character. When covalent bond is formed between two similar atoms, for example in H_2O_2, Cl_2, N_2 or F_2, the shared pair of electrons is equally attracted by the two atoms. As a result electron pair is situated exactly between the two identical nuclei. The bond so formed is called non-polar covalent bond. Contrary to this in case of a heteronuclear molecule like HF, the shared electron pair between the two atoms gets displaced more towards fluorine since the electronegativity of fluorine is far greater than that of hydrogen. The resultant covalent bond is a polar covalent bond. As a result of polarisation, the molecule possesses the **dipole moment** (depicted below) which can be defined as the product of the magnitude of the charge and the distance between the centres of positive and negative charge. It is usually designated by a Greek letter 'i'. Mathematically, it is expressed as follows :

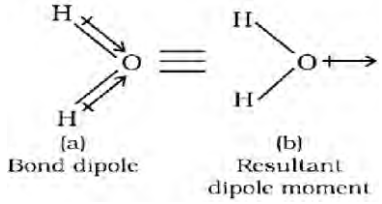
Dipole moment (i) = charge (Q) × distance of separation (r)

Dipole moment is usually expressed in Debye units (D).The conversion factor is 1 D = 3.33564×10^{-30} Cm where C is coulomb and m is meter. Further dipole moment is a vector quantity and is depicted by a small arrow with tail on the positive centre and head pointing towards the negative centre. For example the dipole moment of HF may be represented as :



The shift in electron density is symbolised by crossed arrow (⟶) above the Lewis structure to indicate the direction of the shift. In case of polyatomic molecules the dipole moment not only depend upon the individual dipole moments of bonds known as bond dipoles but also on the spatial arrangement of various bonds in the molecule. In such case, the dipole moment of a molecule is the vector sum of the dipole moments of various bonds.

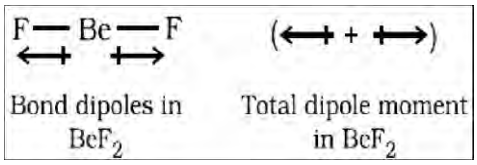
For example in H_2O molecule, which has a bent structure, the two O–H bonds are oriented at an angle of 104.50. Net dipole moment of 6.17×10^{-30} C m (1D = 3.33564×10^{-30} C m) is the resultant of the dipole moments of two O–H bonds.



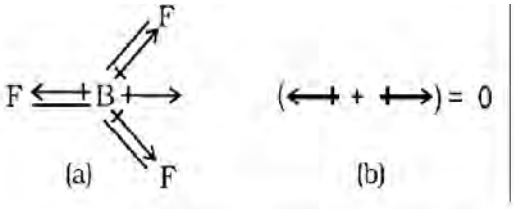
Net Dipole moment, i = 1.85 D

= 1.85 × 3.33564 × 10^{-30} C m = 6.17 ×10^{-30} C m

The dipole moment in case of BeF_2 is zero. This is because the two equal bond dipoles point in opposite directions and cancel the effect of each other.

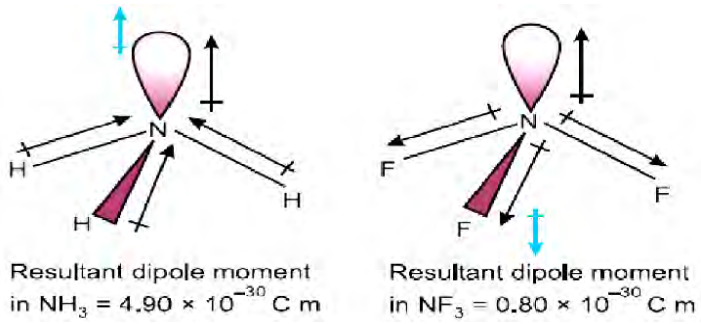


In tetra-atomic molecule, for example in BF_3, the dipole moment is zero although the B – F bonds are oriented at an angle of 120° to one another, the three bond moments give a net sum of zero as the resultant of any two is equal and opposite to the third.



BF_3 molecule; representation of (a) bond dipoles and (b) total dipole moment

Let us study an interesting case of NH_3 and NF_3 molecule. Both the molecules have pyramidal shape with a lone pair of electrons on nitrogen atom. Although fluorine is more electronegative than nitrogen, the resultant dipole moment of NH_3 (4.90×10^{-30} Cm) is greater than that of NF_3 (0.8×10^{-30} C m). This is because, in case of NH_3 the orbital dipole due to lone pair is in the same direction as the resultant dipole moment of the N – H bonds, whereas in NF_3 the orbital dipole is in the direction opposite to the resultant dipole moment of the three N–F bonds. The orbital dipole because of lone pair decreases the effect of the resultant N – F bond moments, which results in the low dipole moment of NF_3 as represented below :



Solved Examples

Example-3 What is the increasing order of ionic character in H_2Se, H_2S, H_2O

(A) $H_2Se < H_2S < H_2O$ (B) $H_2Se > H_2S > H_2O$

(C) $H_2Se < H_2S > H_2O$ (D) None of these

Example-4 Which of the following molecules should not have $\mu = 0$:–

(A) H_2 (B) CO_2 (C) Cl_2 (D) SO_2

Example-5 Which of the following compounds should have higher dipole moment than the remaining three:

(A) HF (B) H_2O (C) NH_3 (D) NF_3

HYDROGEN BOND

This is special case of dipole-dipole interaction. This is found in the molecules in which highly polar N–H, O–H or H–F bonds are present. Although hydrogen bonding is regarded as being limited to N, O and F; but species such as Cl may also participate in hydrogen bonding. Energy of hydrogen bond varies between 10 to 100 kJ mol^{-1}. This is quite a significant amount of energy; therefore, hydrogen bonds are powerful force in determining the structure and properties of many compounds, for example proteins and nucleic acids. Strength of the hydrogen bond is determined by the coulombic interaction between the lone-pair electrons of the electronegative atom of one molecule and the hydrogen atom of other molecule. Following diagram shows the formation of hydrogen bond.

$$\overset{\delta+}{H}-\overset{\delta-}{F}\cdots\overset{\delta+}{H}-\overset{\delta-}{F}$$

Intermolecular forces discussed so far are all attractive. Molecules also exert repulsive forces on one another. When two molecules are brought into close contact with each other, the repulsion between the electron clouds and that between the nuclei of two molecules comes into play. Magnitude of the repulsion rises very rapidly as the distance separating the molecules decreases. This is the reason that liquids and solids are hard to compress. In these states molecules are already in close contact; therefore they resist further compression; as that would result in the increase of repulsive interactions.

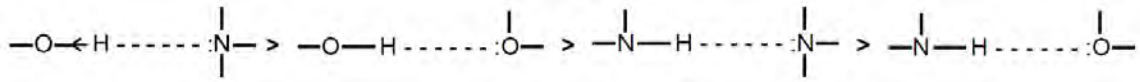
Conditions required for H-bond :

(i) Molecule should have more electronegative atom (F, O, N) linked to H-atom.

(ii) Size of electronegative atom should be smaller.

(iii) A lone pair should be present on electronegative atom.

❍ Order of H-bond strength

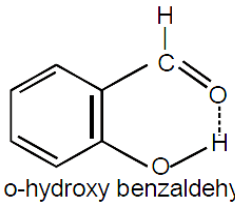


Types of Hydrogen Bonds :

Intramolecular H-Bonding :

This type of H-bonding occurs when polar H and electronegative atom are present in the same molecule i.e., it is formed when hydrogen atom is present in between the two highly electronegative (F, O, N) atoms within the same molecule.

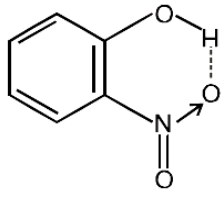
(a)



o-hydroxy benzaldehyde

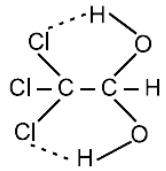
It has lower boiling point (i.e. more volatile) than its para-derivative (where association of molecules takes place using intermolecular H-bonding) because it exists as discrete molecules.

(b)



o-nitro phenol

It has lower boiling point (i.e. more volatile) than its para-derivative (where association of molecules takes place using intermolecular H-bonding) because it exists as discrete molecules.

(c)  Chloral hydrate

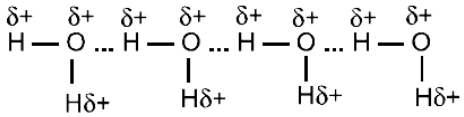
Chloral hydrate is stable only on account of intramolecular hydrogen bonding.

Intermolecular H-Bonding :

Exists between the negative and positive ends of different molecules of the same or different substances i.e., it is formed between two different molecules of the same or different compounds.

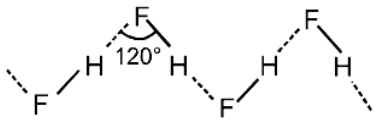
(a) **In water molecules**

Due to polar nature of H_2O there is association of water molecules giving a liquid state of abnormally high boiling point.



- When ice is formed from liquid water, some air gap is formed (in tetrahedral packing of water molecules). The tetrahedral structure around each oxygen atom with two regular bonds to hydrogen and two hydrogen bonds to other molecules requires a very open structure with large spaces between ice molecules. Due to this volume of ice is greater than liquid water and thus ice is lighter than water. We can say that density decreases when ice is formed. Reversely when ice melts, density increases but only upto 4°C, after this intermolecular H-bonding between water molecules breaks hence, volume increases and hence density decreases. Thus, water has maximum density at 4°C.

(b) The hydrogen bonds in HF link the F atom of one molecule with the H-atom of another molecule, thus forming a zig-zag chain $(HF)_n$ in both the solid and also in the liquid.

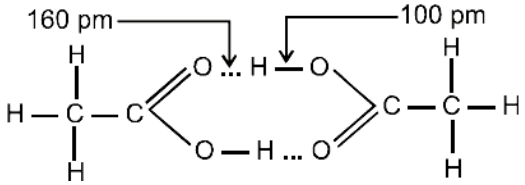


Some hydrogen bonding also occurs in the gas, which consists of a mixture of cyclic $(HF)_6$ polymers, dimeric $(HF)_2$, and monomeric HF.

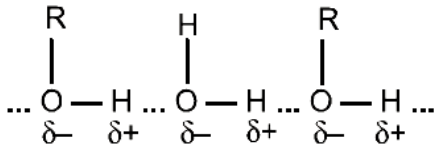
Very strong hydrogen bonding occurs in the alkali metal hydrogen fluorides of formula $M[HF_2]$; in KHF_2, for example, an X-ray diffraction study together with a neutrons diffraction study shows that there is a liner symmetrical anion having an over all, F–H–F distance of 2.26 Å, which may be compared with the H–F bond length of 0.92Å in hydrogen fluoride monomer.

(c) There is also similar H-bonding in alcohol (R—OH) ammonia (NH_3) and phenol (C_6H_5OH) molecules.

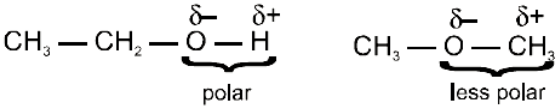
(d) Carboxylic acid dimerises in gaseous state due to H-bonding



(e) Alcohol is said to be highly soluble in water due to crossed intermolecular H-bonding (between H_2O and R—OH molecules).



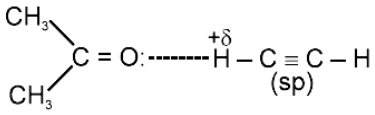
However isomeric ether is less soluble in water due to less polar nature of ether.



Solved Examples

Example-6 C_2H_2 is not soluble in H_2O but it is highly soluble in acetone.

Solution:



In hybridisation as %S character increase, electronegativity increase hence C_2H_2 forms H–bonds with O–atom of acetone and get dissolved. But H_2O molecules are so much associated that it is not possible for C_2H_2 molecules to break that association, hence C_2H_2 is not soluble in H_2O.

Example-7 Explain that tetramethyl ammonium hydroxide is a stronger base than that of trimethyl ammonium hydroxide.

Solution:



In the trimethyl compound the O–H group is hydrogen bonded to Me_3NH group and this makes it more difficult for the OH group to ionize and hence it is a weak base.

In the tetramethyl compound, hydrogen bonding can not occur, so the OH^- group ionizes easily and thus it is a much stronger base.

EXERCISE #1

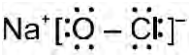
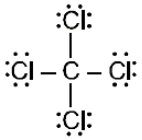
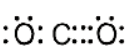
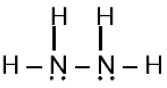
1. Two element have electronegativity of 1.2 and 3.0. Bond formed between them would be :
(A) ionic (B) polar covalent (C) co-ordinate (D) metallic

2. Which forms a crystal of NaCl ?
(A) NaCl molecules (B) Na^+ and Cl^- ions (C) Na and Cl atoms (D) None of these

3. The maximum covalency of representative elements is equal to (excluding 1st and 2nd period) :
(A) the number of unpaired p-electrons
(B) the number of paired d-electrons
(C) the number of unpaired s and p-electrons
(D) the actual number of s and p-electrons in the outermost shell.

4. Which of the following contains both electrovalent, covalent and coordinate bonds?
(A) $MgCl_2$ (B) H_2O (C) NH_4Cl (D) none

5. If the atomic number of element X is 7 the Lewis diagram for the element is :
(A)  (B)  (C)  (D) 

6. Which of the following Lewis diagram is incorrect ?
(A)  (B)  (C)  (D) 

7. The total number of σ and π bonds in $C_2(CN)_4$ are :
(A) 9 σ and 9 π (B) 9 σ and 18π (C) 18 σ and 9π (D) 18 σ and 18π

8. Acetylene consists of :
(A) both sigma and pi bonds (B) sigma bond only
(C) pi bond only (D) none of these

9. The correct order towards bond angle is :
(A) Bond angle does not depend on hybridisation
(B) $sp < sp^2 < sp^3$
(C) $sp^2 < sp < sp^3$
(D) $sp^3 < sp^2 < sp$

10. Which of the following has been arranged in increasing order of % p-character?
(A) $sp < sp^2 < sp^3$ (B) $sp^3 < sp^2 < sp$ (C) $sp^2 < sp^3 < sp$ (D) $sp^2 < sp < sp^3$

11. Which is not true about CH_4 molecule ?
(A) Tetrahedral hybridisation (B) 109.5° bond angle
(C) Four sigma bonds (D) One Ione pair of electrons on carbon

12. Of the following molecules, the one, which has permanent dipole moment, is :
(A) SiF_4 (B) BF_3 (C) PF_3 (D) PF_5

13. The dipole moment of the given molecules are such that :
(A) $BF_3 > NF_3 > NH_3$ (B) $NF_3 > BF_3 > NH_3$ (C) $NH_3 > NF_3 > BF_3$ (D) $NH_3 > BF_3 > NF_3$

14. Pure phosphoric acid is very viscous because :
(A) it is a strong acid
(B) it is tribasic acid
(C) it is hygroscopic
(D) it has PO_4^{3-} groups which are bonded by many hydrogen bonds

15. Which of the following is least volatile ?
(A) HF (B) HCl (C) HBr (D) HI

16. H – bonding is not present in :
(A) NH_3 (B) H_2O (C) H_2S (D) HF

17. Which of the following pair of elements form a compound with maximum ionic character ?
(A) Na and F (B) Cs and F (C) Na and Br (D) Cs and I

18. Which of the following shows the highest lattice energy ?
(A) RbF (B) CsF (C) NaF (D) KF

19. Example of super octet molecule is :
(A) SF_6 (B) PCl_5 (C) IF_7 (D) All of these

20. The number of electrons involved in the bond formation in N_2 molecule is :
(A) 2 (B) 4 (C) 10 (D) 6

21. The octet rule is not obeyed in :
(A) CO_2 (B) BCl_3 (C) PCl_5 (D) (B) and (C)

22. Choose the molecules in which hybridisation occurs in the ground state ?
(1) BCl_3 (2) NH_3 (3) PCl_3 (4) BeF_2
The correct answer is -
(A) 1, 2, 4 (B) 1, 2, 3 (C) 2, 3 (D) 3, 4

23. Which of the following compounds have bond angle as nearly 90º ?
(A) CH_4 (B) CO_2 (C) H_2O (D) SF_6

24. sp^2 – hybridisation is shown by :
(A) $BeCl_2$ (B) BF_3 (C) NH_3 (D) XeF_2

25. The hybridisation of carbon in diamond, graphite and acetylene is –
(A) sp^3, sp^2, sp (B) sp^3, sp, sp^2 (C) sp^2, sp^3, sp (D) sp, sp^3, sp^2

26. Among the following compounds the one that is polar and has central atom with sp^3 hybridisation is:

(A) H_2CO_3 (B) SiF_4 (C) BF_3 (D) $HClO_2$

27. The correct order of dipole moment is :

(A) $CH_4 < NF_3 < NH_3 < H_2O$ (B) $NF_3 < CH_4 < NH_3 < H_2O$

(C) $NH_3 < NF_3 < CH_4 < H_2O$ (D) $H_2O < NH_3 < NF_3 < CH_4$

28. Hydrogen bonding would not affect the boiling point of :

(A) HI (B) NH_3 (C) CH_3OH (D) H_2O

29. Which of the following compound has maximum number of H-bonds per mole ?

(A) HF (B) PH_3 (C) H_2O (D) OF_2

30. Water (H_2O) is liquid while hydrogen sulphide (H_2S) is a gas because :

(A) water has higher molecular weight

(B) hydrogen sulphide is week acid

(C) water molecular associate through hydrogen bonding

(D) sulphur has high electronegativity than oxygen

EXERCISE #2

Weak forces:

1. Which of the following option(s) is/are **CORRECT**?

Type of interaction distance-energy function

(A) dipole-dipole $\propto \frac{1}{r^3}$

(B) H-bonding $\propto \frac{1}{r^3}$

(C) Ion-dipole $\propto \frac{1}{r^2}$

(D) London force $\propto \frac{1}{r^6}$

2. The correct order of the boiling point is/are:

(A) He < Ne < Ar < Kr < Xe

(B) H_2 < He

(C) $H_2 < D_2 < T_2$

(D) $BF_3 < BMe_3$

Bent's and drago's rule:

3. Which of the following order is correct regarding %P character?

(A) H_2S has higher P character in S – H bond than the O – H bond in H_2O

(B) PH_3 has higher P character in P – H bond than the N – H bond in NH_3

(C) NH_4^+ has higher P character in N – H bond than the P–H bond in NH_4^+

(D) None of these

4. Which of the following statement(s) is/are **CORRECT**?

(A) Axial bond length > equatorial bond length in PF_5

(B) axial bond length (P –F) < equatorial bond length (P–Cl) in PF_2Cl_3

(C) All P–F bond lengths are equal in PF_5

(D) axial bond length (P–F) > equatorial bond length (P–Cl) in PF_2Cl_3

5. Choose the correct angle order.

(A) HPH in PH_4^+ = HCH in CH_4

(B) HNH in NH_3 < HPH in PH_3

(C) HNH in NH_3 < HPH in PH_4^+

(D) OSO in SO_3^{2-} < ONO in NO_3^-

6. In which of the following back bonding is **NOT** possible:

(A) $N(CH_3)_3$ (B) BO_3^{-3} (C) $P(CH_3)_3$ (D) BF_4^-

Back bonding:

7. Molecule(s) in which maximum number of atoms in a plane may be ten:

(A) $B_2Me_4H_2$ (B) $Al_2(CH_3)_6$ (C) $N(SiH_3)_3$ (D) $[Co(NH_3)_6]^{3+}$

8. Which of the following statement(s) is/are **CORRECT**:

(A) $[B_3O_6]^{-3}$ ion is non-planar but $B_3N_3H_6$ is planar

(B) $(H_3Si)_3N$ is planar but $(SiH_3)_3P$ is pyramidal

(C) H_3C–N C S is bent but SiH_3 - NCS is linear

(D) $(CH_3)_3N$ is pyramidal but $(GeH_3)_3N$ is planar

9. Which of the following statements is/are **INCORRECT**:

(A) CHF_3 is less acidic than $CHCl_3$

(B) $R_3C–O–H$ is more acidic than $R_3Si–O–H$

(C) In BF_3 back bonding is possible but in CO back bonding is not possible

(D) PH_3 is more basic than NH_3

10. Select the correct order of bond angle.

(A) $O(SiH_3)_2 > OCl_2$ (B) $N(SiH_3)_3 > N(CH_3)_3$

(C) $O(SiH_3)_2 < OCl_2$ (D) $N(SiH_3)_3 < N(CH_3)_3$

11. In which of the following compounds, underlined atom can change their hybridisation due to back bonding:

(A) $\underline{B}_3N_3H_6$ (B) $\underline{N}(SiH_3)_3$ (C) $O(\underline{Si}H_3)_2$ (D) $H_3B\underline{O}_3$

Multicentred bond:

12. No X–X bond exists in which of the following compounds having general form of X_2H_6?

(A) B_2H_6 (B) C_2H_6 (C) Al_2H_6 (D) Si_2H_6

13. Which of the following molecules have **CORRECT** indicated overlapping.

Molecule	**Overlapping in the bridge bond (if any)**
(A) Si_2Cl_6	$sp^3–p–sp^3$
(B) Be_2H_4	$sp^2–s–sp^2$
(C) Si_2H_6	$sp^3–s–sp^3$
(D) B_2H_6	$sp^3–s–sp^3$

Silicate:

14. In which of the following cases the number of corner shared per tetrahedron is '2':

(A) Pyroxene chain silicate (B) Amphibole chain silicate

(C) 5-membered cyclic silicate (D) None of these

Odd electron species:

15. Select correct about NO_2:

(A) It is odd electron specie (B) N–O bond order = 1.5

(C) Paramagnetic specie (D) Isoelectronic with CO_2

16. The number of specie(s) which are not perfectly planar.

(A) $\dot{C}H_3$ (B) $\dot{C}F_3$ (C) $\dot{C}HF_2$ (D) $\dot{C}HF_3$

17. Which of the following statement is **CORRECT**:

(A) The free electron of ClO_3 molecule is present in d-orbital of Cl-atom

(B) The free electron of is $C\dot{F}_3$ present in sp^3 hybrid orbital

(C) NO is polar

(D) The free electron of ClO_2 molecule is present in d-orbital of Cl-atom

Hydrolysis:

18. Which of the following compounds do not give free halogen acid (Hydra acid) on hydrolysis with excess water as a final product?

(A) NCl_3 (B) PCl_3 (C) $SiCl_4$ (D) BF_3

19. Select the **CORRECT** statement(s) re grading the hydrolysis of SF_4:

(A) In transition state H_2O attacks from axial position

(B) Six moles of NaOH are required for the complete neutralisation of hydrolysis product of one mole of compound

(C) Oxidation state of 'S' in product is +6

(D) All the product(s) can form H-bonding

20. Which of the following compounds can form H_3PO_3 on hydrolysis:

(A) P_4O_6 (B) P_4O_7 (C) P_4O_8 (D) P_4O_9

Molecule does not exist:

21. Which of the following species do not exist in nature.

(A) PI_5(TBP form) (B) PbI_4 (C) HFO_3 (D) ICl_7

22. Which of the following do/does not exist?

(A) SH_6 (B) HFO_4 (C) FeI_3 (D) $HClO_3$

Inert pair effect:

23. Which of the following have (18 + 2) electron configuration?

(A) Pb^{2+} (B) Cd^{2+} (C) Bi^{3+} (D) SO_4^{2-}

24. Which of following stability order is/are correct due to inert pair effect.

(A) $Hg > Hg^{2+}$ (B) $Bi^{3+} < Bi^{5+}$ (C) $Pb^{2+} > Pb^{4+}$ (D) $Fe^{2+} < Fe^{3+}$

Ionic compound:

25. Which of the following order is/are **CORRECT**:

(A) $MgCO_3 < BaCO_3$ (Thermal stability) (B) LiF < CsF (Solubility)

(C) $Li_3N > K_3N$ (Thermal stability) (D) $MgSO_3 > BaSO_3$ (Solubility)

26. Select the **CORRECT** order against the mentioned property:

(A) $NaNO_3 < KNO_3 < RbNO_3$ (Thermal stability)

(B) NaF> KF > RbF (Covalent nature)

(C) Si – O < P – O < S – O < C l – O (π-bond strength)

(D) $F_2 < Cl_2 < Br_2 < I_2$ (Bond length)

27. Correct order of solubility in water will be?

(A) LiCl > NaCl > KCl < RbCl < CsCl (B) $Li_2CO_3 < Na_2CO_3 < K_2CO_3$

(C) LiF < NaF < KF < RbF (D) $BeCO_3 < MgCO_3 < CaCO_3$

28. Which of the following will give metal oxide on heating?

(A) $CaCO_3$ (B) $AgNO_3$ (C) K_2CO_3 (D) Li_2CO_3

29. Which of the following order is/are **CORRECT**:

(A) $NaCl < LiCl$ (m.p.)
(B) $CaF_2 > CaO$ (lattice energy)
(C) $LiNO_3 < NaNO_3$ (thermal stability)
(D) $Be_3N_2 > Ba_3N_2$ (thermal stability)

Molecular orbital theory:

30. Which of the following have identical bond order?

(A) O_2^{2+} (B) NO^+ (C) CN^- (D) CN^+

31. Assuming that if Hund's rule is violated, then the diamagnetic specie(s) is/are:

(A) B_2 (B) O_2 (C) $NO^{\ominus}$ (D) $O_2^{\oplus}$

32. The paramagnetic molecule(s) which have non fractional bond order:

(A) O_2 (B) $O_2^{\ominus}$ (C) N_2^{2-} (D) B_2

Miscellaneous:

33. Bond angles which are associated with sp^3d^3 hybridization.

(A) 90º (B) 120º (C) 180º (D) 72º

34. Which of following statement is/are **CORRECT** for $ClOF_3$?

(A) In hybridisation central atom uses its d_{xy} orbital
(B) In π-bond formation central atom uses its 'p' orbital
(C) The shape of molecule is see-saw
(D) The molecule is non-planar

35. Which of the following species are planar as well as polar.

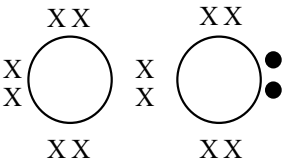
(A) I_3^+ (B) NO_2^- (C) PCl_3 (D) BO_3^{3-}

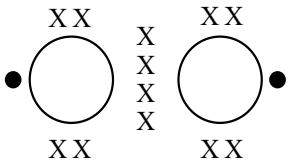
NSEJS PYQ

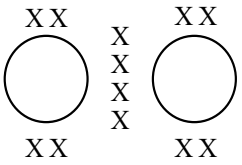
1. Which of the following Lewis dot structures best describes the structure of peroxide ion of sodium peroxide?

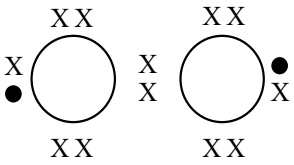
X – electrons from oxygen **[NSEJS 2015]**

• – electrons from sodium

(A) 

(B) 

(C) 

(D) 

2. Elements A, B and C have atomic numbers X, X + 1 and X + 2, respectively. 'C' is an alkali metal. 'A' reacts with another element 'Y' to form the compound 'AY'. 'A' and 'Y' belong to the same group. 'AY' possesses an **[NSEJS 2016]**

(A) ionic bond (B) covalent bond (C) metallic bonding (D) coordinate bond

3. To prevent the formation of oxides, peroxides, and superoxides, alkali metals are sometimes stored in an unreactive atmosphere. Which of the following gases should not be used for lithium: **[NSEJS 2016]**

(A) Ne (B) Ar (C) N_2 (D) Kr

4. An electron pair donor is a Lewis base and an electron pair acceptor is a Lewis acid. Which among the following statements, is correct? **[NSEJS 2016]**

(A) NH_3 is a Lewis acid, because nitrogen has only 6 electrons around it.

(B) BF_3 is a Lewis base, because fluorine has 8 electrons around it.

(C) NF_3 is a Lewis base, because nitrogen has a lone electron.

(D) BCl_3 is a Lewis acid because it has only 6 electrons around it.

5. All of these species have the same number of valence electrons as nitrate ion, except **[NSEJS 2016]**

(A) Carbonate ion (B) Bicarbonate ion (C) NF_3 (D) SO_3

6. Sodium tungstate has formula Na_2WO_4, lead phosphate has formula $Pb_3(PO_4)_2$, formula for lead tungstate should be: **[NSEJS 2019]**

(A) $PbWO_4$ (B) $Pb_2(WO_4)_3$ (C) $Pb_3(WO_4)_2$ (D) $Pb_3(WO_4)_4$

7. The compound/s that raise/s the temperature of water (from room temperature) on dissolving in it is/are- **[NSEJS 2021]**

(A) Ammonium chloride (B) Potassium hydroxide

(C) Glucose (D) Conc. HCl

8. Inter-particle distance between Li and H in LiH is 1.596Å. Observed dipole moment of LiH is 1.964×10^{-29} C.m. The percentage (%) ionic character in LiH is: **[NSEJS 2022]**

(A) 56.0% (B) 90.8% (C) 76.8% (D) 100%

9. H_2 reacts faster with Cl_2 at 13 times faster rate than D_2 because: **[NSEJS 2023]**

(A) H_2 has high activation energy

(B) In H_2, H – H bond energy is higher than D – D bond energy in D_2

(C) H_2 has low activation energy because H – H bond energy is lower than D – D bond energy

(D) In H_2 there is no neutron therefore it reacts faster

10. Select the correct order of dielectric constant, refractive index and intermolecular forces for water (H_2O) and heavy water (D_2O) at 293 K respectively among those given below

(i) Dielectric constant – $H_2O > D_2O$

(ii) Dielectric constant – $D_2O > H_2O$

(iii) Refractive index – $H_2O > D_2O$

(iv) Refractive index – $D_2O > H_2O$

(v) Intermolecular forces – $H_2O > D_2O$

(vi) Intermolecular forces – $D_2O > H_2O$

The option containing all correct statements is **[NSEJS 2023]**

(A) (i), (iii), (vi) (B) (i), (iv), (v) (C) (ii), (iii), (v) (D) (i), (iv), (vi)

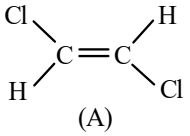
11. The total number of lone pairs of electrons in I_3^-. **[NSEJS 2023]**

(A) 3 (B) 6 (C) 2 (D) 9

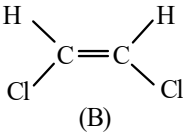
12. One structural isomer of $C_2H_4I_2$ contains both the Iodine (I) atoms attached to same carbon atom. The bond length of C – I bond is 2.10 Å. Assuming tetrahedral angles and additivity of covalent bond radii, the distance between two Iodine (I) atoms is (given that sin 54.75° = 0.815 and sin 109.5° = 0.94) **[NSEJS 2024]**

(A) 2.10 Å (B) 2.47 Å (C) 4.20 Å (D) 3.42 Å

13. The expected values of the normal boiling point of the two compounds A and B given below are TA and TB respectively. Then **[NSEJS 2024]**



(A)



(B)

(A) $T_A < T_B$ (B) $T_A = T_B$ (C) $T_A > T_B$ (D) Can't be predicted

Advanced & Olympiad Challenger Questions

1. The pair in which the first compound is ionic and the second compound is covalent, is:
(A) $Fe(OH)_2$, CH_3OH (B) $Fe(OH)_2$, $Cu(OH)_2$ **[KVPY part-1/2009]**
(C) CH_3OH, CH_3CH_2OH (D) $Ca(OH)_2$, $Cu(OH)_2$

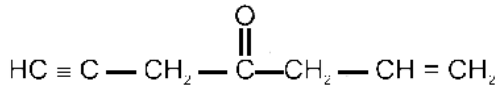
2. Among NH_3, BCl_3, Cl_2 and N_2 the compound that does not satisfy the octet rule is: **[KVPY part-1/2010]**
(A) NH_3 (B) BCl_3 (C) Cl_2 (D) N_2

3. The number of P-H bond(s) in H_3PO_2, H_3PO_3 and H_3PO_4, respectively, is **[KVPY part-1/2011]**
(A) 2, 0, 1 (B) 1, 1, 1 (C) 2, 0, 0 (D) 2, 1, 0

4. The molecule which does not exhibit strong hydrogen bonding is : **[KVPY part-1/2012]**
(A) methyl amine (B) acetic acid (C) diethyl ether (D) glucose

5. Which of the following molecules has no dipole moment ? **[KVPY part-1/2013]**
(A) CH_3Cl (B) $CHCl_3$ (C) CH_2Cl_2 (D) CCl_4

6. The number of sp_2 hybridized carbon atoms in **[KVPY part-1/2013]**



(A) 3 (B) 5 (C) 4 (D) 6

7. The numbers of lone pairs and bond pairs in hydrazine are, respectively **[KVPY part-1/2014]**
(A) 2 and 4 (B) 2 and 6 (C) 2 and 5 (D) 1 and 5

8. Hybridisation of xenon atom in XeF_4 is : **[KVPY part-2/2018]**
(A) sp^3 (B) dsp^2 (C) sp^3d^2 (D) d^2sp^3

9. Among the following mixtures, dipole-dipole as the major interaction is present in: **[IJSO stage-1/2008-09]**
(A) benzene and ethanol (B) acetonitrile and acetone
(C) KCl and water (D) benzene and carbon tetrachloride

10. Hydrogen fluoride is a liquid at room temperature due to : **[IJSO stage-1/2011-12]**
(A) dimerisation (B) dissociation followed by aggregation
(C) association (D) polymerization

11. The property which is characteristic of an electrovalent compound is that **[IJSO stage-1/2011-12]**
(A) it is easily vapourised (B) it has a high melting point
(C) it is a weak electrolyte (D) it often exists as a liquid

12. Which of the following elements forms polyatomic molecules ? **[IJSO stage-1/2011-12]**
(A) Nitrogen (B) Chlorine (C) Argon (D) Boron

13. The compound which contains both ionic and covalent bonds is **[IJSO stage-1/2012-13]**
(A) KCI (B) CS_2 (C) C_2H_6 (D) KCN

14. Which species below has the same general shape as NH_3 ? **[IJSO stage-1/2013-14]**

(A) SO_3^{2-} (B) CO_3^{2-} (C) NO_3^- (D) SO_3

15. What occurs when $H_2O(l)$ evaporates **[IJSO stage-1/2015-16]**

(i) Covalent bonds are broken.

(ii) Only dipole-dipole forces are overcome.

(iii) Heat is absorbed by water from the surroundings.

(iv) It becomes oxygen and hydrogen gas.

(A) Only (i) and (iii) occurs (B) Only (ii) and (iii) occurs

(C) (i), (iii) and (iv) occurs (D) (i), (ii), (iii) and (iv) take place

16. Greenhouse gases absorb (and trap) outgoing infrared radiation (heat) from Earth and contribute to global warming. A molecular that acts as a greenhouse gas, generally has a permanent dipole moment and sometimes for other reasons. Going only by the condition of permanent dipole moment, in the list of gases given below, how many can be potential greenhouse gases?

Water, Sulphur dioxide, Boron trifluoride, Carbon monoxide, Carbon dioxide, Nitrogen, Oxygen, Methane hydrogen sulphite, ammonia. **[IJSO stage-1/2016-17]**

(A) Five (B) Six (C) Seven (D) Four

17. All of these species have the same number of valence electrons as nitrate ion, except **[IJSO stage-1/2016-17]**

(A) Carbonate ion (B) Bicarbonate ion (C) NF_3 (D) SO_3

18. An electron pair donor is a Lewis base and an electron pair acceptor is a Lewis acid. Which among the following statements, is correct?

(A) NH_3 is a Lewis acid, because nitrogen has only 6 electrons around it **[IJSO stage-1/2016-17]**

(B) BF_3 is a Lewis base, because fluorine has 8 electrons around it.

(C) NF_3 is a Lewis base, because nitrogen has a lone electron.

(D) BCl_3 is a Lewis acid because it has only 6 electrons around it

19. The maximum number of Hydrogen bonds in which hydrogen fluoride molecule can participate are : **[IJSO-Stage-II/ 2011]**

(A) 2 (B) 3 (C) 4 (D) 5

20. Which of the following sequence is correct in terms of the polarity of bond **[IJSO-Stage-II/ 2011]**

(A) N-F > C-F > B-F (B) B-F > C-F > N-F (C) C-F > N-F > B-F (D) B-F= C-F > N-F

21. If HCl molecule is completely ionic the H^+ and Cl^- ions would bear a unit charge equal to 4.80×10^{-10} esu and the bond distance between H and Cl atom is 1.27 Aº then the dipole moment of HCl is: **[IJSO-Stage-II/ 2011]**

(A) 3.779 D (B) 7.742 D (C) 6.096 D (D) 2.976 D

22. Liquid state of He is due to **[IJSO-Stage-II/ 2012]**

(A) dipole-dipole interaction (B) ion-dipole interaction

(C) dipole-induce dipole interaction (D) dispersion forces

Answer Key

EXERCISE #1

1. (B)	**2.** (B)	**3.** (D)	**4.** (C)
5. (C)	**6.** (C)	**7.** (A)	**8.** (A)
9. (D)	**10.** (B)	**11.** (D)	**12.** (C)
13. (C)	**14.** (D)	**15.** (A)	**16.** (C)
17. (B)	**18.** (C)	**19.** (D)	**20.** (D)
21. (D)	**22** (C)	**23.** (D)	**24.** (B)
25. (A)	**26.** (D)	**27.** (A)	**28.** (A)
29. (C)	**30.** (C)		

EXERCISE #2

1. (A, B, C, D)	**2.** (A, C, D)	**3.** (A, B)	**4.** (A, B)
5. (A, C, D)	**6.** (A, C, D)	**7.** (A, B)	**8.** (B, C, D)
9. (B, C, D)	**10.** (A, B)	**11.** (B, D)	**12.** (A, C)
13. (B, D)	**14.** (A, C)	**15.** (A, B, C)	**16.** (B, C)
17. (B, C, D)	**18.** (A, D)	**19.** (B, D)	**20.** (A, B, C, D)
21. (A, B, C, D)	**22.** (A, B, C)	**23.** (A, C)	**24.** (A, C)
25. (A, B, C, D)	**26.** (A, B, C, D)	**27.** (A, B, C)	**28.** (A, D)
29. (C, D)	**30.** (A, B, C)	**31.** (A, B, C)	**32.** (A, C, D)
33. (A, C, D)	**34.** (C, D)	**35.** (A, B)	

NSEJS PYQ

1. (D)	**2.** (B)	**3.** (C)	**4.** (D)
5. (C)	**6.** (A)	**7.** (B, D)	**8.** (C)
9. (C)	**10.** (A)	**11.** (D)	**12.** (D)
13. (A)			

Advanced & Olympiad Challenger Questions

1. (A)	**2.** (B)	**3.** (D)	**4.** (C)
5. (D)	**6.** (A)	**7.** (B)	**8.** (C)
9. (B)	**10.** (C)	**11.** (B)	**12.** (D)
13. (D)	**14.** (A)	**15.** (B)	**16.** (A)
17. (C)	**18.** (D)	**19.** (A)	**20.** (B)
21. (C)	**22.** (D)		